

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
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(1) Research Program/Project/Study Title: : Nanohydroxyapatite from oyster shells for agri-aquaculture applications	
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Academic Rank:	Asst. Prof. 1
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Pontevedra						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☒ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☒ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Fisheries/Nanomaterials		(8) Discipline/Field of Specialization Nanotechnology							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

In the Philippine archipelago, several islands are excellent for growing oysters. A sizable portion of the nation's production of oysters is produced across 17 provinces, including Capiz in Region 6. Capiz ranks first in oyster production in Region 6 in 2020. (PSA, 2020). Oysters are popularly consumed in households, as a best-selling dish in restaurants, and as a tourist attraction. There is no data on the volume of oyster shell waste, but commercial production and consumption account for the generation of solid waste from oyster shells. Essentially, the edible portion of oyster shellfish is 20-30% of its total weight, thus, in Capiz alone, roughly 6,400 MT of oyster shell waste is generated in 2020 (based on the volume of production in PSA Report 2020).

Problems concerning health and environment may arise from accumulation of these waste. However, they can be repurposed as viable source of nanomaterials which can be utilized for agri-aquaculture applications. Hydroxyapatite nanomaterials can be used as starting materials for fertilizer and feed formulation and as bio-adsorbents of pollutants in aquaculture farms, applications which are very useful in the university, as well as in the province and the country. This project focuses on the synthesis of valuable nanomaterials, specifically, nanohydroxyapatite from oyster shell byproducts which will be anchored on green nanotechnology.

These plans align with the HNRDA-AANR's support to 1) Agriculture 4.0 that is smart, green and S&T-based" especially in the frontier technologies such as nanotechnologies; 2) high-value product development from agricultural wastes and 3) technology commercialization initiatives.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

The province of Capiz has rich oyster farms, providing for commercial and household consumers. Tourists coming to the province look forward to having a taste of the flavorful meat of the oyster, locally known as "talaba". The shells of oysters are generally discarded as waste, which adds to the volume of municipal solid waste. Additionally, poor waste disposal might leave the dumping location as a niche for bacterial and fungal growth and may leave a foul smell in the neighboring areas. Recycling of shell waste is highly encouraged to reduce its volume. One way to do this is to transform these byproducts into high-value bio-based materials.

With this, Capiz State University is one with the University of the Philippines Visayas in promoting their registered trademark, AVAMOS, which stands for *A Valorisation of Agri-Fisheries Materials using Opportune Science*. Through this partnership, CapSU will explore possibilities for conducting research that adheres to the concepts of green chemistry and circular economy in order to produce high-value materials from waste oyster products to be utilized in agri-aquaculture applications.

Taking into account UPV's capabilities to support nanotechnology research and the accessibility of raw materials at CapSU, this project aims to optimize the methods of nano-hydroxyapatite synthesis and characterization for agri-aqua industry applications. Specifically, it has the following objectives:

1. synthesize nano-hydroxyapatite from oyster shells using low-temperature heating and green solvents
2. characterize the synthesized nano-hydroxyapatite from oyster shells using
3. optimize synthesis method in terms of:
 - a. temperature
 - b. pH
 - c. rate of addition of the phosphate source
4. use oyster shells as nanoadsorbents for the removal and/or recovery of excess nitrogen and

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phosphorus from aquaculture effluents

The outlined tasks hope to demonstrate the power of interdisciplinary collaboration in ensuring that innovation based on green nanotechnology processes and the circular economy has a positive impact on society. The nanohydroxyapatite synthesized from this study will be used in developing bio-adsorbents for aquaculture ponds.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The overall goal of this project is to utilize oyster shell byproducts as a source of useful nanoparticles for agri-aqua industries applications. Generally, this project aims to optimize the methods of nano-hydroxyapatite synthesis and characterization using green chemistry for agri-aqua industries applications.

Specific:

1. synthesize nano-hydroxyapatite from oyster shells using low-temperature heating and green solvents
2. characterize the synthesized nano-hydroxyapatite from oyster shells
3. optimize synthesis method in terms of:
 - a. temperature
 - b. pH
 - c. rate of addition of phosphate source
4. use oyster shells as nanoadsorbents for the removal and/or recovery of excess nitrogen and phosphorus from aquaculture effluent

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Oyster Production in the Philippines

Fisheries production is crucial for the economy of the Philippines since it generates stable jobs and money. In particular, there are various islands in the archipelago of the Philippines that are ideal for oyster farming. The regions I, IV, and VI are home to the 17 oyster farm provinces. The majority of oysters are produced in particular in the provinces of Pangasinan and Cavite. The Philippines' Bulacan and Cavite regions, which are near to Metro Manila City, the nation's capital, produced over 28% of the nation's total oyster output (5.9 million tons annually) between 2015 and 2017. A significant portion of the local fishery sector is made up of raw oysters and byproducts. The livelihood of the local populations is strongly dependent on the rising oyster and mussel farming industries in the Philippines because they require little capital investment and only modest labor skills; FAO has proposed that the Philippines has greater potential industries (Chilakala et al., 2019).

Calcium compounds from shells

The shells from processed bivalves are discarded into marine waters. This is harmful as it can destroy the vegetation and habitat of marine organisms in the area where they are dumped. Rather than discard shells, converting them into a more valuable product can greatly benefit the environment and the economy. It reduces wastes that go into landfills or bodies of water and thus decreases our carbon footprint. It also generates a product with potentially high market value – e.g. the calcium compound, hydroxyapatite, that can be derived from bivalves shells are projected to reach a global market value of 3.6B by 2027 (<https://www.alliedmarketresearch.com/hydroxyapatite-market>). Other potential valorization products of mussel shells include calcium supplements (Xu et al. 2020); potential material for artificial reef construction (Wallis et al. 2016); liming agent in agriculture (Morris, Backeljau, and

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Chapelle 2018); catalysts in biodiesel production (Perea et al. 2016) and biosorbents for the removal of pollutants from water (Du, Lian et al. 2011). Here, the focus will be on the conversion of mussel shells to hydroxyapatite using solvents generally regarded as safe (GRAS).

Nano-hydroxyapatite (n-HAP) - synthesis and applications

In recent years, researchers working in the field of biomaterials have become interested in hydroxyapatite ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, HA), an important inorganic biomaterial. Since hydroxyapatite (HA) closely resembles bone apatite and has good biocompatibility, it is successfully used as a bioimplant material. Due to its chemical and structural similarity with the mineral phase of bone and teeth, HA is widely used for hard tissues repair. As a result, this inorganic phosphate has been studied extensively for medical applications in the form of powders, composites or even coatings (Agrawal et al., 2011). Hydroxyapatite (HAP) is a biomineral well-known for its use in bone and dental implant applications (Piccirillo, Pullar et al. 2015), bioceramic for biomedical applications (Sunil and Jagannatham 2016), ion-exchange and sorption capacity applications (Goto and Sasaki 2016), and scaffold fabrication (Mondal, Mondal et al. 2016) due to its porosity, size, and same chemical properties to the mineral component of bones and hard tissues of living mammals, specifically humans (Nam, Hoa et al. 2019). HAP can be recovered from bones by a simple treatment (Zainol, Adenan et al. 2019). Here, the treatment of mussel shells follows the works of (Ni et al., 2003 and Zaremba et al., 1998) in which shells were reacted with mild reagents (e.g. ammonium biphosphate) to form hydroxyapatite. The functionalization of n-HAP with urea has been discussed by (Kottegoda, Sandaruwan et al. 2017) and will be the basis of the application studies for n-HAP.

In the study of Anjaneyulu, et al (2014), they developed a method which is simple and affordable for the synthesis of nano HAP with large scale quantity for outstanding applications in dental and orthopedic fields. They synthesized their nHAP at a lower temperature by using sol-gel method with two different calcium sources and similar phosphorus material. The formed phase structure and morphology were found to be most affected by pH, temperature, pressure, MW power, and duration. Additionally, it has been demonstrated that the morphology and size of the produced nHA are influenced by the feed Ca/P molar ratio, the chelating effect of EDTA, and the synergistic effect of ultrasound. The preparation of nHA with the help of MW was discovered to be remarkably effective and affordable. MW-assisted methods have the advantages of selective and effective super heating due to the optional/changeable MW radiation frequency, holding time and power that lead to the extra fast reaction process. Additionally, the "sintering" of biomaterials during MW processing could help with the intra-operative processing and application of HA bone grafts, which are being considered for a variety of applications to replace and repair bone tissues. Additionally, research is being done on additional contentious and unstudied issues, such as the thermal versus non-thermal effects of MW-assisted nHA synthesis (Hassan et al., 2016).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested).

Hydroxyapatite (HA) is the most frequently used calcium phosphate-based biomaterials and it could be prepared from natural or synthesized sources via several processes (Hassan, et al., 2016). Shells have been explored as viable sources of calcium-based products which are used for agricultural functions like soil amendments, feed additives, and the like. Most of the researches on hydroxyapatite nanoparticles are centered on biomedical applications (Ahmed et al., 2022; Ishikawa et al., 2020). This study focuses on the utilization of the byproducts from oyster production as raw materials for the synthesis of hydroxyapatite nanomaterials which can be essentially used for agri-aquaculture functions.

Figure 1 shows the interrelated processes in the project. Oyster production is a sustainable food source; it is consumed either in its fresh or processed form. This entails the generation of shell waste which can cause environmental hazards in the long run. However, solid waste accumulation may be reduced by repurposing them as useful biomaterials for various applications.

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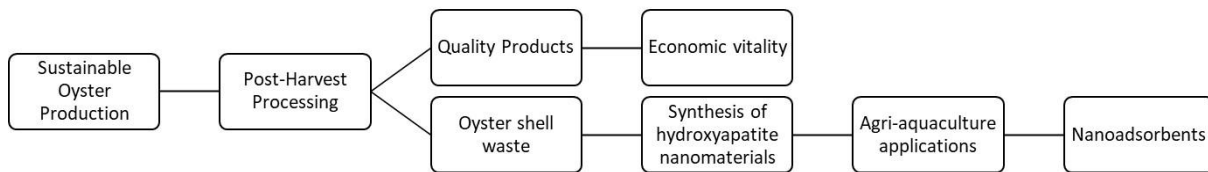


Figure 1. Schematic diagram of interrelated activities in this project

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Oyster shell collection and preparation

Oyster shells will be collected from the Sangha Aqua-Adventure Farm of Capiz State University, Pontevedra, Capiz and will be used as feedstock for the synthesis of nanohydroxyapatite for agriculture and aquaculture applications. The shells will be cleaned using the shell washer fabricated by the Agricultural and Biosystems Engineering Department of Capiz State University. It will be rinsed with distilled water and sun-dried or oven dried until it is dry enough to be pulverized using the shell mill from the same department. It will then be saved for further processing, characterization and synthesis.

Synthesis, functionalization and characterization of Nanohydroxyapatite

Microwave-assisted Hydrothermal Synthesis

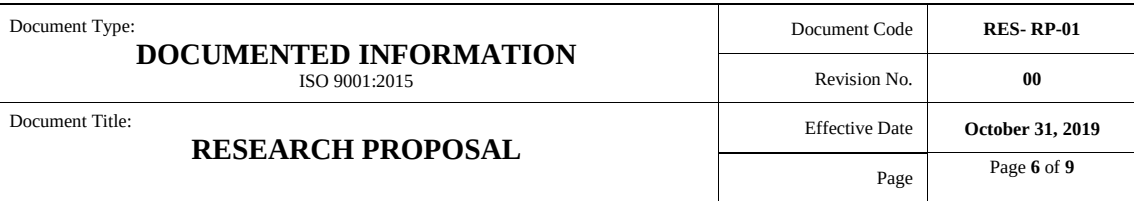
Nano- hydroxyapatite (nHAP) will be produced from waste oyster shells using green nanotechnology methods. One is microwave-assisted hydrothermal synthesis as outlined by (Shavandi, Bekhit et al. 2015). Oyster shells will be pretreated with a chelating agent, EDTA, for thirty minutes then placed in a microwaveable Teflon lined hydrothermal vessel and irradiated with a microwave with a power of 1.1 kW. The effect of irradiation time and power will be determined. The resulting products will be filtered, washed and dried for further analysis. Chemical composition, size, structure and thermal stability of the prepared n-Hap will be characterized using SEM, XRD, FTIR and TGA. The n-HAP will be functionalized by combining the works of Shvandi, Bekhit et al. (2015) and Kottegoda, Sandaruwan et al. (2017).

Sol-gel Synthesis

Oyster shell will be pretreated acetic acid. The calcium acetate and orthophosphoric acid will then be taken and dissolved in double-distilled H₂O. To maintain Ca/P ratio 1.67, orthophosphoric acid will be added slowly, under vigorous stirring, to calcium acetate. Alcohol will be added into the homogenous water solution of calcium acetate and phosphorus alkoxide. The necessary quantity of alcohol for gelation decreases in the order Methyl, Ethyl, Propyl alcohol. After gel formation, drying follows at 75~ for a few days. The dried gel will be thermally treated up to 930°C (Cihlar and Castkova, 1998; Kordas and Trapalis, 1997). The resulting products will be subjected to further analysis.

Wet chemical precipitate Method

The oyster shell powder will be bleached for 30 minutes. The bleached oyster shell powder will be treated with acetic acid to form calcium acetate will be added with 2.5 mL ammonium hydroxide, subjected to ultrasound (50W) for 1 hour. This will be done at pH 8, 9 and 10. Potassium hydrogen phosphate will be added at instantaneous and slow rates and will be subjected to ultrasound for 1 hour (50 W). At this point, a white precipitate is expected to be formed, which will then be filtered and calcined at (T= 300 deg C, 600 deg C and 900 deg C). Resulting products will be subjected to further



Instrumentation for the Characterization of the Nanomaterials

1. Fourier Transform Infrared (FTIR) Spectroscopy – to identify the functional groups in hydroxyapatite. FTIR, is often used in conjunction with other structural methods of analysis such as XRD, XPS, NMR, MS to determine structure and identity of compounds
2. Scanning Electron Microscopy (SEM) of hydroxyapatite – to illustrate macro- and microstructure images at different magnifications; to evaluate particle/grain size and geometry
3. Energy Dispersive Spectroscopy (EDS) of HAp – for elemental characterization and molar ratio (Ca/P) analysis
4. Atomic Force Microscopy (AFM) – this is necessary to perform sub-atomic resolution particle sizing, topography scanning and determine nanosize-dependent properties of hydroxyapatite
5. X-ray Diffraction (XRD) of hydroxyapatite – to determine the crystal structure of the hydroxyapatite to allow for comparison with structure of hydroxyapatite in bones and teeth
6. Thermogravimetric Analysis (TGA) – to determine the chemical composition of HAp with respect to temperature
7. Atomic Absorption Spectroscopy (AAS) – for heavy metal characterization and molar ratio (Ca/P) analysis

This study aims to optimize the method of synthesis of nanohydroxyapatite using response surface methodology (RSM) as the experimental design.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.)

(6) OBJECTIVES	(7) TARGET ACTIVITIES	(8) TARGET ACCOMPLISHMENTS (quantify, if possible)	YEAR 1				YEAR 2			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1. synthesize nano-hydroxyapatite from oyster shells using low-temperature heating and green solvents	1.1 hiring and training of personnel and procurement of supplies and reagents 1.2 Training of personnel on project management, laboratory safety and use of high-tech instrumentation 1.3 collection and preparation of kg quantities of pulverized oyster shells 1.4 synthesis of nanohydroxyapatite from oyster shell	1.1.1 Two (2) PAs 1.2 4 project personnel trained 2 trainings conducted 1.2.1 Supplies and reagents procured 1.3.1 Powdered oyster shell samples (minimum of 10 kg) 1.4.1 Nanohydroxyapatite powder	x	x						
				4 2						
			x	x	x	x				
			x	x						
				x	x					
2. characterize the synthesized nano-hydroxyapatite from oyster shells	2.1 characterization of the synthesized nanohydroxyapatite from oyster shells 2.2 functionalization with P and N and characterization of n-HAP with N,P	2.1 Technical specification sheet of nanohydroxyapatite powder 2.2.1 Nanohydroxyapatite (n-HAP) powders Technical specification sheet of n-HAP powders			x	x	x	x		
3. optimize synthesis method in terms of: 3.1 temperature 3.2 pH 3.3 rate of addition of the phosphate source	3.1 statistical analysis of data to determine the optimum temperature, pH and rate of addition of the phosphate source	3.1.1 optimized method of synthesis 1 Manuscript draft 1 UM application 1 trademark application 1 BS/MS mentored			x	x	x	x		
4. use oyster shells as nano-adsorbents for the removal and/or recovery of excess nitrogen and phosphorus from aquaculture effluent	4.1 analysis of nitrogen and phosphorus adsorption on nanohydroxyapatite	4.1.1 adsorption isotherms 1 Manuscript draft 1 UM application 1 trademark application 1 BS/MS mentored					x	x	x	1 1

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(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Publication

- at least two (2) original articles on the product development and application, economic valuation oyster shell waste utilization
- at least two (2) abstracts in conferences
- at least two (2) posters
- at least two (2) technical manuals/guides
- at least two (2) technical specifications
- one training module

Patent/Intellectual Property

- at least two (2) IP (UM) applications – method of synthesis hydroxyapatite nanoparticles and process of producing nanohydroxyapatite adsorbents
- at least two (2) trademarks applied

Product

- at least two (2) powders (nano-adsorbent and hydroxyapatite nanoparticle raw material) as products with potential for commercialization.
- at least two (2) processes - process of synthesizing hydroxyapatite nanoparticles and process of producing nanohydroxyapatite adsorbents
- one (1) training program

People Service

- at least one (1) training conducted
- at least three (3) young faculty and research personnel trained
- at least two (2) BS or MS students graduated
- (2) Invention disclosure report
- (2) prior art search report
- (1) technology pricing report
- (1) IP valuation report
- (1) Technology valuation report
- (1) technical feasibility report
- (1) financial feasibility report
- (1) market research report

Place and Partnership

- one (1) partnership (with UPV) strengthened
- one (1) partnership forged (e.g. with an aquaculture farm operator)

Policy

at least one policy input on the integrated use of oyster shell products to support the creation of livelihoods

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Oyster producers and processors
Nano-adsorbent developers

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Pond operators and farmers
Researchers

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
KRIS B. ROXAS	Project Leader	Chemistry
CONCEPCION P. PONCE	Project Staff	Nanotechnology
RICARDO BABARAN	Project Staff	Fisheries
REY J. DELA CALZADA	Project Staff	Aquaculture

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

IMPLEMENTING AGENCY	PS	MOOE	EO	TOTAL
Year 1	1,001,472.00	1,176,095.80	300,000.00	2,477,567.80
Year 2	1,001,472.00	876,095.80		1,877,567.80
Year n				
TOTAL				4,355,135.60

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)

Ryan, Lucy. <https://thumbs.dreamstime.com/b/pile-discarded-oyster-shells-close-up-whitstable-beachfront-kent-199301598.jpg>